



Credo

Proof-triggered RWA settlement for Creditcoin.

Official USDC stays final on Ethereum. The ERC-1155 stays on Creditcoin. Only a cryptographic proof crosses chains — and the contract, not the backend, decides.

Attestcoin Protocol · USC v2

Creditcoin CC3 Testnet

Ethereum Sepolia

THE PROBLEM

Cross-chain RWA settlement is broken in four ways.

- **Bridged money is not the seller's money.** Wrapped USDC introduces custody, depeg and bridge-risk into a payment that was already final.
- **Oracles assert; they don't prove.** A backend saying "payment happened" is a promise. Settlement deserves cryptographic evidence.
- **Operators hold god-mode.** Most pipelines can redirect or fake settlement — a compromise away from losing every asset.
- **Payments are loosely coupled to delivery.** Without binding buyer, token, amount and time-window on-chain, one transfer can settle the wrong deal.

THE SOLUTION

Delivery-versus-payment, proven on-chain.

PAYMENT STAYS PUT

The buyer signs one plain USDC transfer on Sepolia. No bridge message, no wrapper, no second network for the user.

PROOF CROSSES

Attestcoin turns the Sepolia receipt into a Merkle inclusion + continuity proof. That evidence is the only thing that crosses.

CONTRACT DECIDES

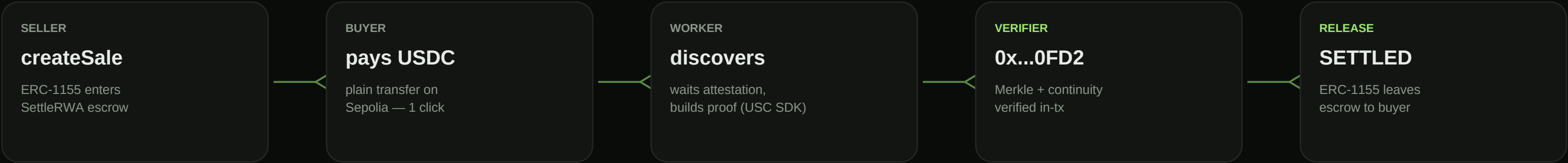
SettleRWA releases escrow only when the verifier precompile accepts the proof and every bound field matches — synchronously, in one transaction.

The result feels like a normal payment for the buyer — and settles like a derivative contract for the protocol.

MECHANISM

One payment. One proof. One release.

ATTESTCOIN PROTOCOL — only this crosses the chains



ETHEREUM SEPOLIA — payment stays here, final

CREDITCOIN CC3 — asset stays here, final

The worker discovers. Attestcoin proves. The contract decides. The blockchain owns the state.

CORE SCORING

The proof is the only path to settlement.

- **Synchronous in-transaction verification.** `executePaymentProof` calls `verifyAndEmit` on precompile `0x...0FD2` inside the settlement tx — failure reverts everything.
- **Receipt decoded from verified bytes.** Official `EvmV1Decoder` extracts status, emitter, payer, recipient, amount from the attested transaction — never worker claims.
- **Exact-match policy.** Exactly one USDC Transfer must match all bound fields; zero or multiple matches revert.
- **chainKey $\neq$ chainId.** Attestcoin key 1 (Sepolia) validated independently from EVM id 11155111.
- **Source block window.** Payment height must fall in the sale's $\leq 50,000$ -block window — old payments are structurally dead.
- **Global replay guard.** `queryId = keccak(chainKey, height, txIndex)` is one-use; marker + release share one atomic transaction.
- **Official SDK in the worker.** `@gluwa/usc-sdk` `ProofBuilder` + `PrecompileChainInfoProvider` against the CC3 proof builder.

TRUST BOUNDARY

Nobody — including us — can fake a settlement.

What the contract enforces

- Checks-effects-interactions + ReentrancyGuard
- Non-upgradeable, custom errors only
- One OPEN sale per exact payment tuple
- Release and replay marker in one tx
- Reclaim only after window + 24 h grace

What the operator cannot do

- Force-settle or redirect escrowed assets
- Mark arbitrary proofs as verified
- Change a sale's buyer after creation
- Cancel a sale after the buyer paid
- Touch funds — worker key holds no roles

Honest limitation: the Sepolia payment is irreversible before the Creditcoin release completes — this is proof-triggered settlement, not an atomic two-chain swap. The reclaim grace period is the mitigation until reverse write-back ships on the Attestcoin roadmap.

EVIDENCE

Settled live on CC3 — with two independent wallets.

489 s

payment → settlement
latency, end-to-end

2

independent wallets
(distinct seller & buyer)

1 / 0

buyer ERC-1155 balance /
escrow balance, after

true

global replay marker
set in settlement tx

Buyer pays 5 USDC on Sepolia
sepolia.etherscan.io/tx/0xfd25089e189c53a5a39fc81eb0054e1069f66e3a761ddedeb64cd64df2383010

Attestcoin proof accepted (queryId 0x4c94e8b1...)
recorded in docs/evidence/attestcoin/proof-run.json

Settlement executes on Creditcoin
creditcoin-testnet.blockscout.com/tx/0xcf9dd953498e5378d4f815b997a6a1a44e8c03ee6a3ebf23f1c4b3c2956eff96

Every number on this page was re-verified directly against CC3 RPC during the final audit.

BUILDER-READY

Integrate the layer. Keep your product.

- **In-app documentation.** Quickstart, API reference, parameter spec, ABI downloads and security model served at /infra.
- **Compiler-generated ABIs.** SettleRWA · PaymentVerifierUSC · TestRWA — byte-identical to the deployed artifacts, downloadable.
- **Six-step quickstart.** Authenticate → approve → prepare → createSale → index → register payment. Everything after payment is automatic.
- **Strict API contract.** Zod-validated JSON, SIWE-style wallet auth, HttpOnly sessions, typed error codes with request IDs.
- **25 automated tests + 7 Foundry tests** including a 256-run fuzz suite on the settlement invariants.

SETTLEMENT LAYER

Protocol owns truth. Ecosystem owns experience.

CREDO PROTOCOL (this repo)

- Escrow + deterministic release
- Attestcoin proof verification
- Payment-intent uniqueness
- Asset adapter standard (roadmap)
- Worker automation

ECOSYSTEM (builders invited)

- Marketplaces & listing UIs
- OTC desks
- Issuance & compliance platforms
- Discovery, pricing, analytics
- Any RWA app needing DvP

Any application can settle an RWA through the same escrow: approve the contract, call createSale, and the payment-to-release pipeline runs automatically.

FROM PRIMITIVE TO STANDARD

The layer grows without changing the core.

NOW	Settlement primitive Live on CC3: escrow, proof, release, replay guard — verified end-to-end.	● SHIPPED & VERIFIED
PHASE B	Reservation layer Listing → buyer reserves → unique payment intent. Kills the double-buyer race; opens multi-buyer.	
PHASE C	Asset adapters + issuance IAssetAdapter for ERC-20/721/3643; compliance hooks at the adapter layer.	
PHASE D	Discovery / marketplace Index over reservations — escrow stays in SettleRWA; venues never hold assets.	
PHASE E	Atomic two-way DvP When Attestcoin ships reverse write-back, refund legs become trustless.	

WHAT THIS IS NOT

Stated plainly, because trust is the product.

- **Not an atomic two-chain swap.** Payment is irreversible before release completes; the 24 h reclaim grace covers the proof period.
- **Not production value.** Testnet-only deployment; Blockscout source verification is pending; curated asset onboarding.
- **No open marketplace yet.** By design (ADR-009): private sales eliminate the double-buyer race until reservations exist.
- **Not permissionless onboarding.** Asset allowlisting is operator-managed today; governance path is roadmap work.



Move ownership with verifiable payment.

No bridge. No wrapped USDC. No trusted worker.

github.com/<repo>/credo

Live app: this repo, `npm run dev`

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